04. Lightcone bootstrap with controlled large-spin remainders

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A quantum correlation can be calculated by grouping the participating excitations in different ways, and all valid groupings must agree. That agreement can force some excitations to exist and restrict their properties. This problem studies the pattern at large spin, where useful approximations become available. The central challenge is to say not only what the leading approximation predicts, but also how large its error could be.

Key terms

- **Correlation function** — A quantity describing how field measurements at different spacetime points are related.
- **Four-point function** — A correlation involving four field insertions; it can be reorganized into different pairings.
- **Bootstrap** — A method that constrains a theory by requiring its own consistency rules to agree, rather than starting from a complete microscopic model.
- **Scaling dimension Δ** — A measure of how a field or excitation changes when all lengths are rescaled.
- **Spin ℓ** — A label describing how an excitation transforms under rotations.
- **Lightcone limit** — A limit in which two spacetime points approach a separation that light could traverse; it organizes which contributions dominate the correlation function.
- **Twist** — The combination $\Delta-\ell$, useful for ordering contributions near a lightlike separation.
- **Operator product expansion (OPE)** — A rule expressing the effect of nearby field insertions as a sum of other operators with specified coefficients.
- **Double-twist family** — A sequence of large-spin operators whose dimensions approach a characteristic pattern related to two external operators.
- **Remainder bound** — A guaranteed limit on the terms omitted from an approximation.

Why this matters

Consistency can reveal features shared by many strongly interacting theories, even when their detailed dynamics cannot be solved. Large-spin patterns are also useful in theories proposed to describe gravity through a lower-dimensional boundary. Controlled errors turn suggestive asymptotic patterns into statements that can actually constrain a finite calculation.

What progress would mean

The baseline would reproduce an established large-spin pattern with unambiguous conventions. A stronger result would show how far that pattern can deviate at specified spins, making later comparisons and exclusions more trustworthy.

Core question

For one scalar four-point function in a unitary CFT with d>2, what controlled bounds on large-spin double-twist data follow from specified low-twist exchanges?

Scope and assumptions

Start in d=3 with an identical scalar ϕ of fixed dimension $\Delta\phi$. Specify the normalization of C_T, the stress-tensor OPE coefficient fixed by its Ward identity, and an explicit twist gap above the low-twist exchanges retained. Use a large-C_T expansion only when a factorization assumption and its error order are stated.

This challenge studies analytic large-spin behavior. Challenge 07 instead studies the spinning stress-tensor correlator. Torus modular invariance is restricted to a separate d=2 variant.

Mathematical target

For a fixed scalar block convention, define $F_{\Delta,\ell}=v^{\{\Delta\phi\}}g_{\Delta,\ell}(v,u)-u^{\{\Delta\phi\}}g_{\Delta,\ell}(v,u)$, and use $F_{id}+\Sigma\,\lambda^2F=0$. Twist is $\tau=\Delta-\ell$. The double-twist families have $\Delta_{n,\ell}=2\Delta\phi+2n+\ell+\gamma_{n,\ell}$ asymptotically.

Choose n=0 initially and derive a retained contribution *$\hat{\gamma}${0,ℓ}, together with a bound* $|\gamma\{0,\ell\}-\hat{\gamma}_{0,\ell}|\le R(\ell)$ for $\ell\ge\ell_0$ under explicit spectral/Regge assumptions. If only an asymptotic series is obtained, label it as such and do not manufacture a finite-spin bound.

Required outputs and proof obligations

Export block normalizations, low-twist input, the derivation or inversion formula, coefficients, and a decomposition of R into omitted exchanges, expansion terms, and numerical errors. State clearly whether the result is an identity, a conditional bound, or an asymptotic estimate.

Validation and rejection controls

Use generalized-free-field crossing as an algebraic benchmark, while noting that it is not by itself the target local CFT with a finite stress tensor. Recover double-twist accumulation in a known analytic example. Verify every uniform estimate on the specified cross-ratio domain and all spins covered by the claim.

Milestones and research extension

Baseline: reproduce an established leading large-spin result with conventions checked. Strong: obtain one explicit remainder estimate, or identify the extra data needed to obtain it. Research extension: a tighter conditional bound on low-twist exchanges or bulk couplings with a stated holographic dictionary.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Fitzpatrick et al., The Analytic Bootstrap and AdS Superhorizon Locality — Large-spin double-twist structure and analytic bootstrap.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$F_{\Delta,\ell} = v^{\Delta_\phi} g_{\Delta,\ell}(u,v) - u^{\Delta_\phi} g_{\Delta,\ell}(v,u), \qquad F_{\mathrm{id}} + \sum_{\mathcal{O}} \lambda^2_{\phi\phi\mathcal{O}} F_{\Delta,\ell} = 0.$$

$$\Delta_{n,\ell} = 2\Delta_\phi + 2n + \ell + \gamma_{n,\ell}, \qquad |\gamma_{0,\ell} - \hat{\gamma}_{0,\ell}| \le R(\ell) \quad (\ell \ge \ell_0).$$

The inequality is a target requiring explicit spectral and Regge assumptions and a proved remainder. An asymptotic series alone does not imply this finite-spin inequality. Higher-dimensional crossing is not torus modular invariance.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — modular invariance outside its domain (23, 45).** Thermal correlators on S^(d−1)×S^1 do not generally have the torus SL(2,Z) invariance asserted here. The revised d=3 problem uses crossing and Lorentzian analyticity only.
2. **Major — spectrum terminology (16, 31–41).** A large holographic gap normally concerns a selected single-trace sector, not all nonconserved operators; double-twist operators remain. Δ_gap−J_max is not a definition of a minimum twist. The scalar unitarity lower bound is also not a holographic gap prediction.

3. **Major — asymptotics versus finite bounds.** Lightcone expansions alone do not supply certified finite-spin exclusion plots. A remainder or additional input is required.
4. **Major — overlap and overclaim (29, 53–60).** The original mixes analytic scalar bootstrap, spinning bootstrap, modular constraints, and uniqueness of Einstein gravity. The replacement isolates a calculable question and preserves gravity interpretations as conditional extensions.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For one scalar four-point function in a unitary CFT with d>2, what controlled bounds on large-spin double-twist data follow from specified low-twist exchanges?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: reproduce an established leading large-spin result with conventions checked. Strong: obtain one explicit remainder estimate, or identify the extra data needed to obtain it. Research extension: a tighter conditional bound on low-twist exchanges or bulk couplings with a stated holographic dictionary.

Sources supporting the corrections

- Fitzpatrick et al., The Analytic Bootstrap and AdS Superhorizon Locality — Large-spin double-twist structure and analytic bootstrap.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented

trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.

- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.